DEEP LEARNING

Dataset: It's a palmer penguins from kaggle. It's a clasification dataset with 3 species and exactly 4 numerical features, So its fits our 4-2-3 neural network.

Input layer → 4 neurons

X = [39.1, 18.7, 181, 3750]

x_1 = Bill / culmen length

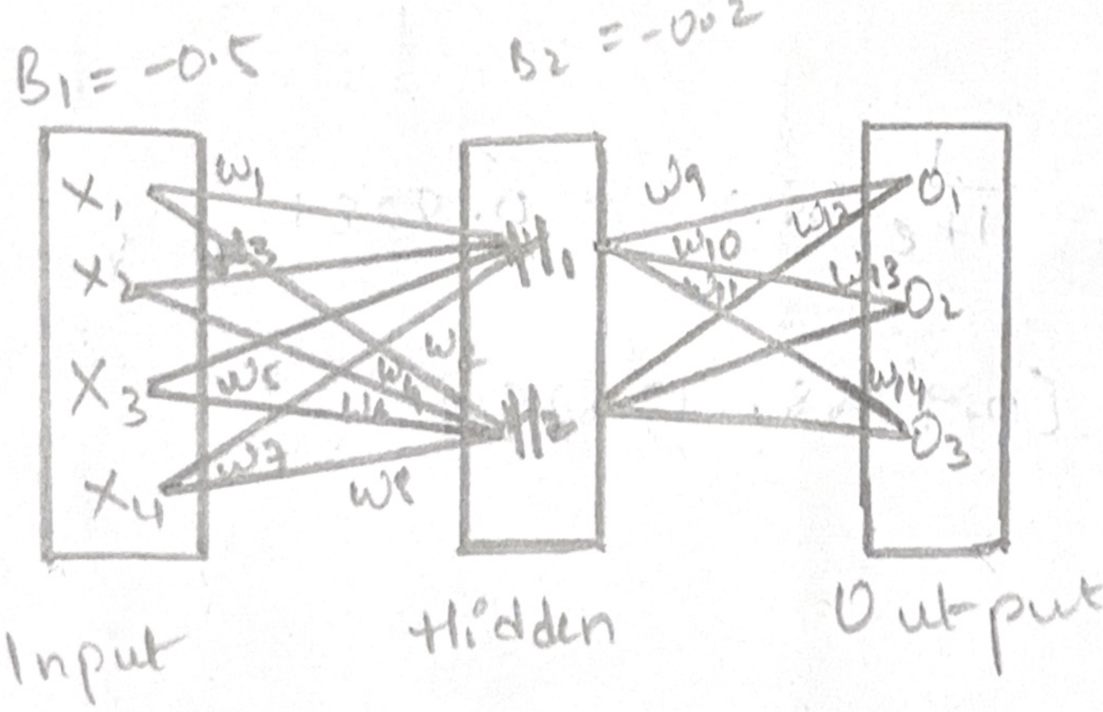
x_2 = Bill / culmen depth

x_3 = flipper length

x_4 = Body mass

Output layer → 3 neurons

Adelie, Chinstrap Gentoo



One Row:

Input = [39.1, 18.7, 181, 3750]

Weights for H_1 = $w_1 = 0.05$, $w_3 = -0.08$, $w_5 = 0.01$, $w_7 = 0.102$

Weights for H_2 = $w_2 = 0.03$, $w_4 = 0.06$, $w_6 = 0.008$, $w_8 = 0.001$

Bias $B_1 = -0.5$

Bias $B_2 = -0.2$

Calculating H_1

$$Z_{H_1} = X_1 w_1 + X_2 w_3 + X_3 w_5 + X_4 w_7 + B_1$$

$$= (39.1 \times 0.05) + (18.7 \times -0.08) + (181 \times 0.01) +$$

$$(3750 \times 0.0002) - 0.5$$

$$= 1.995 - 1.496 + 1.81 + 0.75 - 0.5$$

$$\boxed{Z_{H_1} = 2.519}$$

Sigmoid for H_1

$$H_1 = \frac{1}{1+e^{-2.519}} = \underline{0.9255}$$

Calculating H_2

$$Z_{H_2} = X_1 W_2 + X_2 W_4 + X_3 W_6 + X_4 W_8 + B_2$$

$$= (39.1 \times -0.03) + (18.7 \times 0.06) + (181 \times 0.008) + (3750 \times 0.0001) - 0.2$$

$$= -1.173 + 1.122 + 1.448 + 0.375 - 0.2$$

$$\boxed{Z_{H_2} = 1.572}$$

Sigmoid for H_2

$$H_2 = \frac{1}{1+e^{-1.572}} = 0.8281$$

$$\therefore H = [0.9255, 0.8281]$$

Output layers :

Adelie : $w_9 = 1$, $w_{10} = -0.2$

Chinstrap : $w_{11} = -0.5$, $w_{12} = 1$

Gentoo : $w_{13} = 0.2$, $w_{14} = -0.1$

for Adelie $Z = H_1 w_9 + H_2 w_{10}$

$$= (0.9255 \times 1) + (0.8281 \times -0.2)$$

$$= 0.9255 - 0.1656$$

$$= \underline{0.7599}$$

for Chinstrap $Z = H_1 W_{11} + H_2 W_{12} - 0.1$

$= (0.9255 \times 0.5) + (0.8281 \times 41) - 0.1$

$= -0.4628 + 0.8281 - 0.1$

$= 0.2653$

for Gentoo $Z = H_1 W_{13} + H_2 W_{14} - 0.05$

$= (0.9255 \times 0.2) + (0.8281 \times -0.) - 0.05$

$= 0.1851 - 0.0828 - 0.05$

$= 0.0523$

Output = [0.7599, 0.2653, 0.0523]

Highest Score = 0.7599

∴ Predicted class = Adelie

Final Answer = Adelie